



Interpretable Topic Modelling of Mental Health Conversations in Australian Online Forums Using Network-Based Insights

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Abstract. Mental health conditions such as anxiety, depression, post-traumatic stress disorder (PTSD), and suicidal thoughts significantly impact the wellbeing of many Australians. Beyond Blue, a leading national mental health platform, offers a unique forum where individuals anonymously share experiences and seek support. Compared to global platforms like Reddit and Twitter, this community remains relatively underexplored as a data source for understanding lived experiences of mental illness. In this study, we apply topic modelling and quantitative analysis to Beyond Blue forum posts, identifying key themes and subtopics within categories such as medication, therapy, relationships, and work stress. Using MentalBERT-enhanced BERTopic, we also construct a visual network of subtopics, capturing both diversity and overlap across conditions. The findings reveal the complexity of how Australians articulate distress, cope with challenges, and engage in peer support. This work provides insights to guide tailored interventions, inform digital mental health services, and support policies grounded in lived experience.

Keywords: Beyond Blue · Mental health · Topic modeling · MentalBERT · Machine Learning

1 Introduction

Anxiety, depression, post-traumatic stress disorder (PTSD), and suicidal thoughts are major public health concerns in Australia. Beyond Blue, a leading mental health initiative, provides resources and an anonymous forum where individuals share experiences, seek guidance, and support [1, 2]. These discussions capture first-hand accounts of distress, coping, and peer support, offering a valuable yet underused source of insight into mental health in Australia [3].

While research on online mental health communities has revealed patterns of behaviour, emotion, and support-seeking [4, 5], most work has focused on global platforms such as Reddit and Twitter [6]. Australian-specific forums remain underexplored, and subtopics such as medication, therapy, relationships, and work-related stress have received little attention [1].

This study addresses that gap by applying natural language processing (NLP), particularly BERTopic [7], to Beyond Blue forum data. Leveraging MentalBERT [8, 9] enables more accurate capture of domain-specific language. Our contributions are threefold: (i) identifying recurrent subtopics across anxiety, depression, PTSD, and suicidal thoughts; (ii) constructing a similarity-based network to highlight connections across conditions; and (iii) providing evidence to guide clinicians, policymakers, and digital services in developing more responsive interventions.

2 Methodology

2.1 Dataset Description

This study drew upon publicly accessible discussion threads from the Beyond Blue forums [1], incorporating both initial posts and their corresponding comments (Table 1). For each entry, metadata were retained, including post ID, title, main text, author username, user ID, creation date, forum category, number of comments, and the full text of replies. Out of the ten available forum categories, four were selected for analysis due to their close alignment with key mental health conditions: Anxiety, Depression, PTSD and Trauma, and Suicidal Thoughts and Self-Harm. The selected dataset comprised 2,978 posts on Anxiety, 2,982 on Depression, 2,996 on PTSD and Trauma, and 2,617 on Suicidal Thoughts and Self-Harm. Together, these categories capture a diverse spectrum of lived experiences, while maintaining thematic coherence across conditions.

This structured dataset provides a balanced basis for topic modelling and embedding analysis, ensuring that both individual narratives (posts) and community responses (comments) are represented.

Table 1. Summary of data fields extracted from Beyond Blue forums.

Variable	Description
Post ID	Serves as a unique reference number for each forum entry
Post Title	Heading or subject line provided by the author
Post Content	Main body text of the initial discussion post
Post Author	Display name chosen by the forum contributor
User ID	Distinct identifier allocated to each registered participant
Post Date	The timestamp indicating when the post was created
Post Category	The thematic section of the forum in which the post appeared
No. of Comments	Total number of responses linked to the original post
Comments	Full textual content of all replies within the thread

2.2 Data Preprocessing

Since user-generated forum posts often contain informal language, symbols, and irregular formatting, preprocessing was essential to enhance the quality of topic modelling and

embedding. Two complementary pipelines were developed to balance interpretability with contextual richness.

For topic modelling, posts underwent comprehensive cleaning: removal of irrelevant content using regular expressions, conversion of emojis into text, lowercasing, stop-word removal, lemmatization, tokenization, and n-gram extraction. This process allowed the model to capture domain-specific multiword expressions frequently used in mental health discourse [4, 5].

For embedding-based analysis, only light preprocessing was applied, including lowercasing, emoji conversion, and removal of irrelevant content. This ensured that contextual nuance and emotional tone were preserved—critical for embeddings derived from transformer-based models [6–9]. By combining these two strategies, we produced structured input for clearer topic interpretation while retaining the subtle emotional and linguistic signals that are characteristic of mental health narratives.

2.3 BERTopic

BERTopic is an advanced topic modelling approach that combines transformer-derived embeddings with clustering methods and class-based term weighting to generate coherent and interpretable themes [10]. In contrast to traditional techniques like Latent Dirichlet Allocation, it makes use of dense semantic embeddings and adaptive clustering strategies, enabling the detection of subtle patterns within specialised text corpora [11, 12]. The overall process can be described through six core stages:

1. Embedding. The first stage involves converting forum posts into dense vector representations that capture semantic meaning. For this study, we employed MentalBERT (see Sect. 2.4) as the embedding model, since it is specifically trained to reflect the vocabulary and nuances of mental health language [9].

2. Dimensionality Reduction. Because embeddings are produced in a high-dimensional space, we applied UMAP (Uniform Manifold Approximation and Projection) to compress them into fewer dimensions while retaining local structural relationships [13, 14].

$$z_i \in UMAP(e_i), z_i \in \mathbb{R}^k, k \ll m \quad (2.1)$$

Here, e_i is the embedding of document i in the original m -dimensional space, and z_i is its projection in the reduced k -dimensional space.

The UMAP algorithm represents data as weighted k -nearest neighbor graphs in high and low dimensions, aligning them by minimizing cross-entropy.

$$\mathcal{L}_{UMAP} = \sum_{(i,j)} [w_{ij} \log \sigma(\|z_i - z_j\|) + (1 - w_{ij}) \log(1 - \sigma(\|z_i - z_j\|))] \quad (2.2)$$

where w_{ij} denotes the edge weight (similarity) between points i and j , and σ is the logistic function. As a result, samples that are adjacent in the original embedding space remain positioned near each other after dimensionality reduction.

3. Clustering. After dimensionality reduction, the vectors z_i were organized into groups using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). This algorithm identifies clusters by locating regions with higher data density, adapting automatically to variations in cluster size and shape. Unlike methods that require the number of clusters to be fixed beforehand, HDBSCAN determines this from the data itself. Points that do not strongly belong to any cluster are treated as noise, ensuring that only well-supported groups are retained as topics [15].

4. Vectorizers. To make clusters interpretable, BERTopic applies text vectorization (e.g., CountVectorizer), converting each group into a bag-of-words representation that allows the calculation of keyword importance [16].

5. C-TF-IDF. To highlight words that best characterize each cluster, BERTopic applies a class-based TF-IDF weighting. This method increases the importance of terms that occur frequently within a topic but remain uncommon across the full corpus, thereby improving separation between topics [17, 18].

$$W_{x,c} = \|tf_{x,c}\| \times \log\left(1 + \frac{A}{f_x}\right) \quad (2.3)$$

where $tf_{x,c}$ is count of term x in class c , f_x is the overall frequency of x in the corpus, and A is the average number of documents per class. This reweight emphasizes terms that are frequent in one topic but rare overall.

6. Representation Models and Optimization. To generate clear and non-redundant topic labels, BERTopic employs methods such as Maximal Marginal Relevance (MMR), which balance relevance to the topic with diversity among selected terms [19].

$$MMR = \arg \max_{t \in T \setminus S} \left[\lambda * \text{sim}(t, q) - (1 - \lambda) * \max_{s \in S} \text{sim}(t, s) \right] \quad (2.4)$$

where T is the pool of candidate terms, S is the set of chosen terms, q is the topic representation, $\text{sim}(\cdot)$ is cosine similarity function, and $\lambda \in [0,1]$ controls the trade-off between relevance to the topic and redundancy with selected terms.

To optimize topic quality, the BERTopic pipeline was tuned using a stochastic grid search across UMAP, HDBSCAN, and CountVectorizer parameters [10]. For CountVectorizer, `min_df` was fixed at 1 to hold valuable and meaningful terms, `max_df` was tuned to filter overly frequent words, and unigrams to trigrams were tested alongside an expanded stop-word list. UMAP was set with 15 neighbors, 5 components, and `min_dist` = 0.0 to balance structure and produce compact clusters. HDBSCAN parameters included `min_cluster_size` to control the smallest topic size and `min_samples` to adjust noise sensitivity, with higher values yielding more conservative clusters [11, 14]. This careful tuning ensured the resulting topics were both interpretable and reflective of the variability in mental health discourse.

2.4 MentalBERT

To enhance BERTopic, we employed MentalBERT, an extension of BERT model on mental health-related texts, primarily from Reddit [6, 8]. Unlike general-purpose models,

it is tuned to capture the vocabulary, linguistic cues, and emotional tone of mental health discourse, making it well suited for Beyond Blue forum analysis.

MentalBERT retains the standard BERT architecture [9, 10], combining token, segment, and positional embeddings contextualized through multi-head self-attention. To obtain sentence-level embeddings required by BERTopic, we applied mean pooling across token embeddings using the SentenceTransformers framework [12]. These embeddings were then fed into the BERTopic pipeline Sect. 2.3, ensuring that extracted topics more accurately reflected the lived experiences and language of forum users.

2.5 Verification and Naming New Topics

After extracting topics with BERTopic, we assessed their quality and assigned clear labels. This combined quantitative evaluation with human-in-the-loop review to ensure statistical validity and semantic clarity. Two metrics were used:

Coherence Score: Indicates the degree of semantic alignment among a topic's leading words, where higher scores correspond to more meaningful topics [7, 12].

$$C = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} \cos_sim(w_i, w_j) \quad (2.5)$$

Topic Diversity: Indicates the degree to which topics differ from one another by calculating the share of distinct terms across all topic keyword lists. Higher diversity values show that topics cover a wider range of themes with less repetition [12].

$$D = \frac{\text{Count of distinct terms in all topic lists}}{\text{Total number of terms across all topic lists}} \quad (2.6)$$

In addition, topics were manually reviewed for interpretability and labeled based on representative terms. A large language model (LLM) supported this step by refining labels and checking their semantic fit [14, 17, 18]. This human-in-the-loop process ensured that the final topic names were both technically sound and reflective of forum discourse.

3 Experimental Results and Discussion

3.1 Data Analysis

Data were collected by scraping 150 pages for each of the four Beyond Blue categories: Anxiety, Depression, PTSD and Trauma, and Suicidal Thoughts and Self-Harm. Since earlier posts were sparse and inconsistent, the analysis focuses on the period from 2017 to July 2025, ensuring sufficient data coverage across all categories. Since data collection concluded in mid-2025, the figures for that year reflect only a partial dataset.

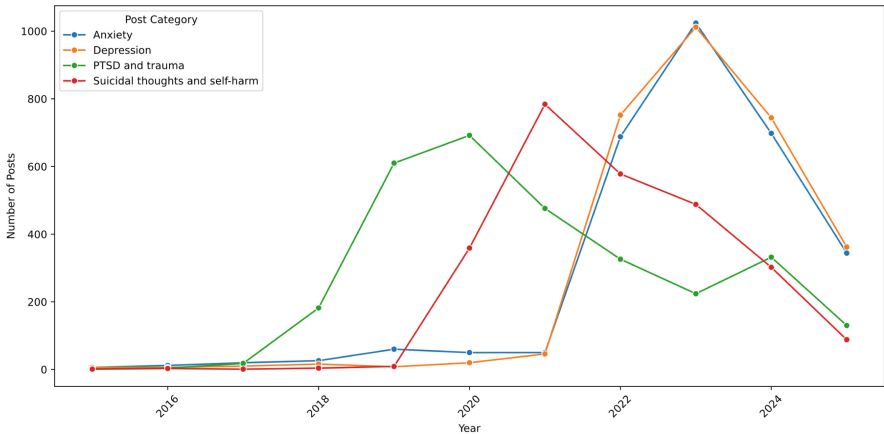


Fig. 1. Trends in reported mental health conditions over the last ten years.

From Fig. 1, posts on anxiety and depression steadily increased, peaking in 2023 before a slight decline, yet they continue to dominate discussions. PTSD- and suicide-related posts appeared in smaller but stable volumes, indicating consistent representation in the forums. These trends provide context for the topic modelling analysis by highlighting temporal variation in mental health discourse and underscoring the central role of anxiety and depression in participants’ lived experiences.

3.2 Hyperparameter Tuning Analysis

To optimize topic quality, a randomized search was conducted over predefined parameter grids for UMAP, HDBSCAN, and CountVectorizer. This approach facilitated exploration across a broad range of parameter values without requiring exhaustive trials. Each setting was assessed with coherence and diversity metrics, offering a combined view of topic clarity and distinctiveness.

Table 2. Tuned BERTopic parameters across mental health categories

Category	max_df	min_cluster_size	min_samples	Coherence	Diversity
Anxiety	0.5	60	40	0.469	0.777
Depression	0.5	60	40	0.507	0.728
PTSD	0.5	55	35	0.455	0.646
Suicide thought	0.45	55	39	0.451	0.600

The optimal settings for the four categories are summarized in Table 2. Results show that Anxiety and Depression achieved the highest coherence scores (0.469 and 0.507, respectively), reflecting clearer and more semantically consistent clusters. In contrast, PTSD and Suicide-related discussions had lower coherence and diversity values, suggesting greater heterogeneity in the way participants articulated these conditions.

Overall, the tuned parameters produced a balance between cluster compactness and topical coverage, ensuring that the extracted topics were both interpretable and sensitive to domain-specific variations.

3.3 Analysis of Topic Modelling Results

In this section, the Anxiety category is presented as a case study, selected because of its prominence in the dataset (Fig. 1) and the wide range of subtopics it generated. BERTopic produced several clusters, which were further refined using an LLM-assisted relabeling process to generate concise and interpretable topic names.

Figure 2 shows the topic word score visualizations, highlighting the most representative keywords that define each subtopic.

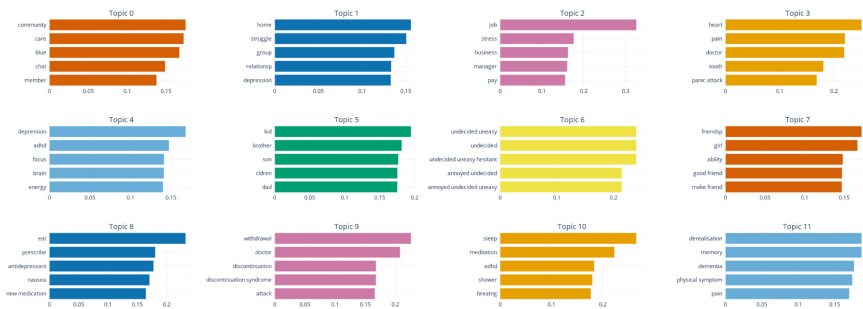


Fig. 2. Topic words score for Anxiety subtopics

Table 3 lists the extracted subtopics with their labels, top keywords, and document counts. Outliers detected by HDBSCAN were excluded for clarity.

Preliminary results show that Anxiety-related discussions cluster around themes such as medication, therapy, workplace stress, and interpersonal relationships. These findings highlight the multifaceted nature of anxiety, encompassing both clinical aspects (e.g., treatment and therapy) and broader social dimensions (e.g., work and relationships). This case study demonstrates how BERTopic with MentalBERT embeddings can reveal fine-grained, semantically meaningful subtopics beyond broad diagnostic categories, offering deeper insights into mental health discourse in Australia.

3.4 Network Visualization

To explore relationships between subtopics, we constructed a semantic similarity network of topics across all four mental health categories. As shown in Fig. 3, each node corresponds to a subtopic, with node size proportional to the number of forum posts assigned to it. Edges indicate semantic similarity between subtopics, where thickness reflects cosine similarity of their MentalBERT embeddings.

This visualization provides an intuitive map of how discussions cluster within and across conditions. Several densely connected regions emerged, such as subtopics relating to medication and therapy, which often co-occur across Anxiety and Depression.

Table 3. (continued)

Model-generated topics	Assigned label	Top keywords	Number of posts
Topic 3	Panic Symptoms	'heart', 'pain', 'doctor', 'tooth', 'panic attack', 'chest pain', 'palpitation', 'sleep', 'physical symptom', 'anxious'	215
Topic 4	ADHD Medication	'depression', 'adhd', 'focus', 'brain', 'energy', 'skill', 'medication', 'stress', 'emotion', 'master'	156
Topic 5	Family Issues	'kid', 'brother', 'son', 'cldren', 'dad', 'mother', 'father', 'relationsp', 'cld', 'old'	138
Topic 6	Uncertainty Skepticism	'undecided uneasy', 'undecided', 'undecided uneasy hesitant', 'annoyed undecided', 'annoyed undecided uneasy', 'skeptical annoyed undecided', 'annoy undecided uneasy', 'skeptical annoy undecided', 'annoy undecided', 'hypnotherapy'	135
Topic 7	Friendship Struggles	'friendsp', 'girl', 'ability', 'good friend', 'make friend', 'guilt', 'hate', 'gh school', 'bully', 'uni'	104
Topic 8	Medication Effects	'eat', 'prescribe', 'antidepressant', 'nausea', 'new medication', 'dosage', 'psycatrist', 'biscuit', 'apricit', 'panic attack'	92

(continued)

Table 3. (continued)

Model-generated topics	Assigned label	Top keywords	Number of posts
Topic 9	Withdrawal Challenges	'withdrawal', 'doctor', 'discontinuation', 'discontinuation syndrome', 'attack', 'pain', 'antidepressant', 'concern', 'reflux', 'breating'	84
Topic 10	Sleep Coping	'sleep', 'meditation', 'adhd', 'shower', 'breating', 'exercise', 'sleep story', 'essential oil', 'night', 'impost'	77
Topic 11	Cognitive Symptoms	'derealisation', 'memory', 'dementia', 'physical symptom', 'pain', 'anxious', 'migraine', 'medication', 'breating', 'tms'	67

By examining this structure, we gain insight into the shared and unique aspects of mental health discourse. Strongly connected clusters point to common challenges that cut across diagnostic categories, while isolated nodes highlight more condition-specific or stigmatized issues. Together, the network underscores the complexity of lived experiences and provides a foundation for identifying areas where targeted interventions or tailored digital support services may be most beneficial.

4 Conclusion

This study analyzed Beyond Blue forum posts to uncover fine-grained themes across Anxiety, Depression, PTSD, and Suicidal Thoughts. Using BERTopic with MentalBERT embeddings, we identified subtopics spanning both clinical concerns (e.g., medication, therapy) and social challenges (e.g., relationships, workplace stress). A semantic network further revealed shared issues across conditions as well as isolated, condition-specific themes. Our findings show that Beyond Blue is a valuable yet underused source for understanding Australian mental health discourse. By combining domain-specific embeddings with topic modelling, we demonstrated a method capable of producing nuanced, interpretable insights that can inform targeted interventions and digital support services.

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